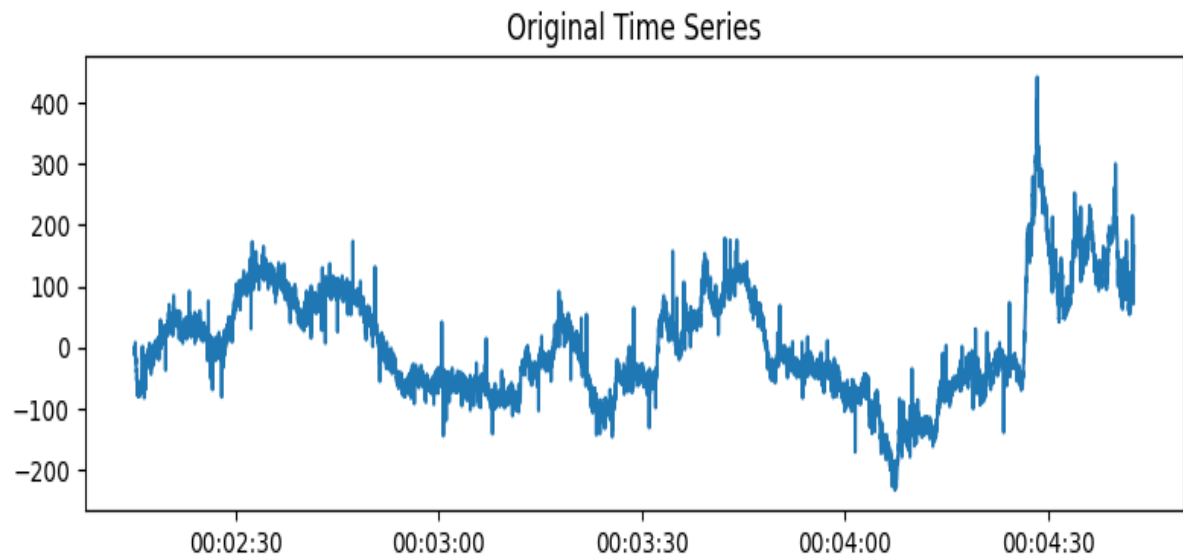


Final Time Series EDA Report

Original Time Series Plot

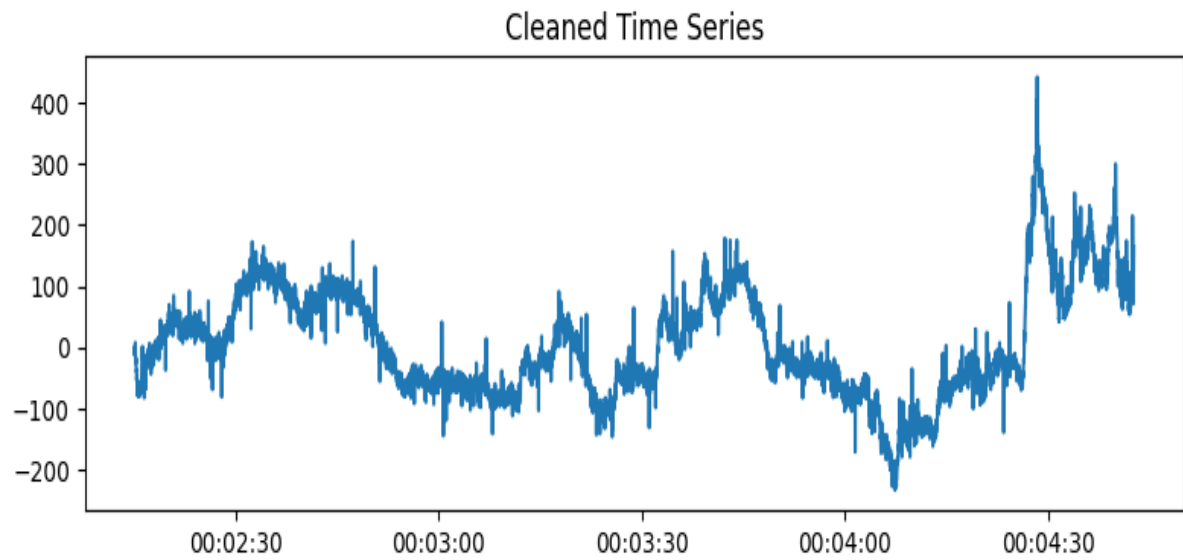


Context & Interpretation: Visualizes raw corrected red signal over time. Oscillatory behaviour indicates physiological signal dynamics.

Python Script:

```
plt.plot(df['timestamp'], df['red_corrected'])
```

Cleaned Time Series Plot

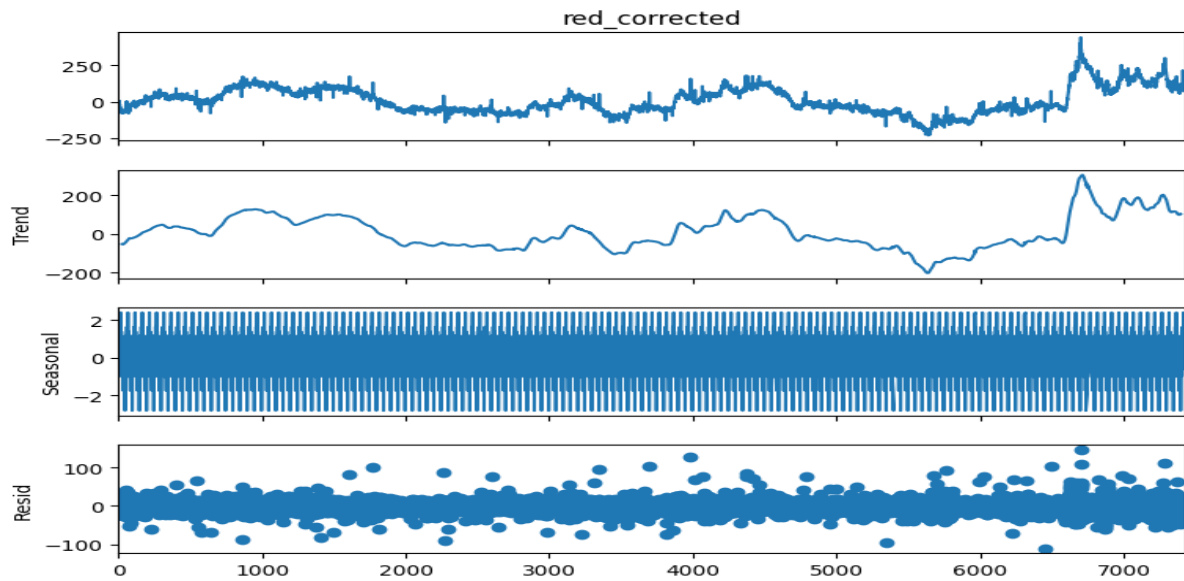


Context & Interpretation: No missing values were present; cleaned series remains unchanged.

Python Script:

```
plt.plot(clean_df['timestamp'], clean_df['red_corrected'])
```

Decomposition Plot

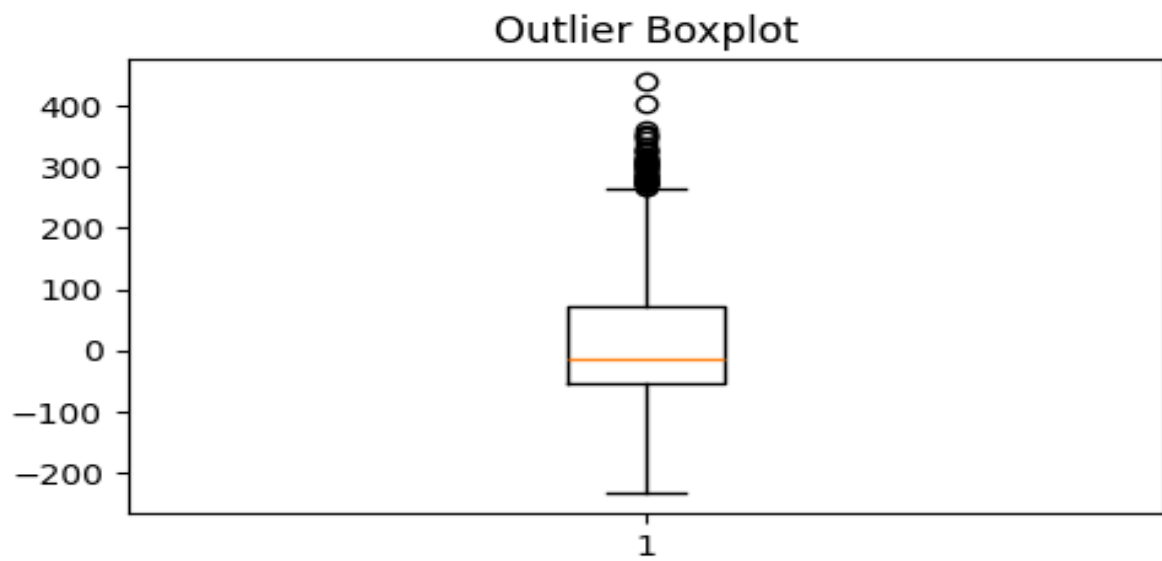


Context & Interpretation: Separates trend, seasonal and residual components for temporal structure analysis.

Python Script:

```
seasonal_decompose(df['red_corrected'], period=50)
```

Outlier Boxplot

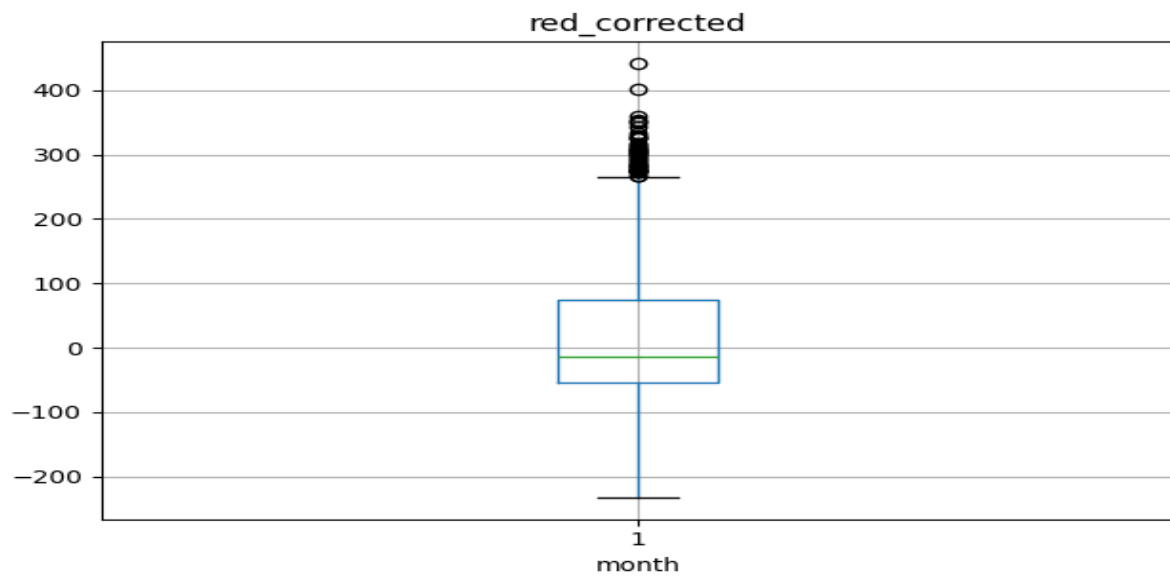


Context & Interpretation: Used to identify extreme observations using IQR-based visualization.

Python Script:

```
plt.boxplot(df['red_corrected'])
```

Monthly Boxplot

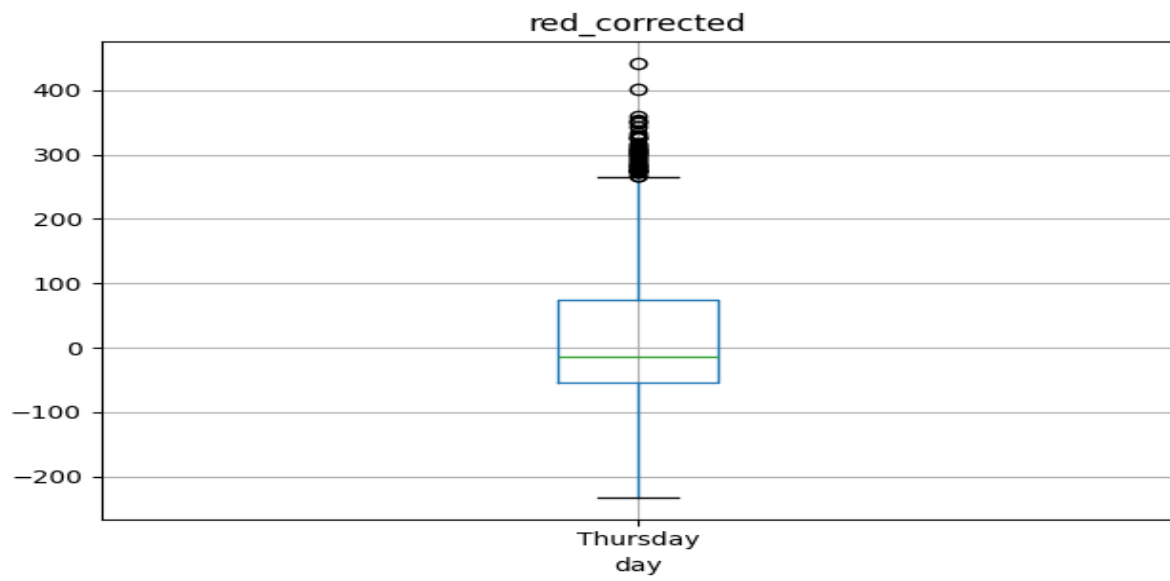


Context & Interpretation: Limited usefulness because dataset spans only minutes.

Python Script:

```
df.boxplot(column='red_corrected', by='month')
```

Weekly Boxplot

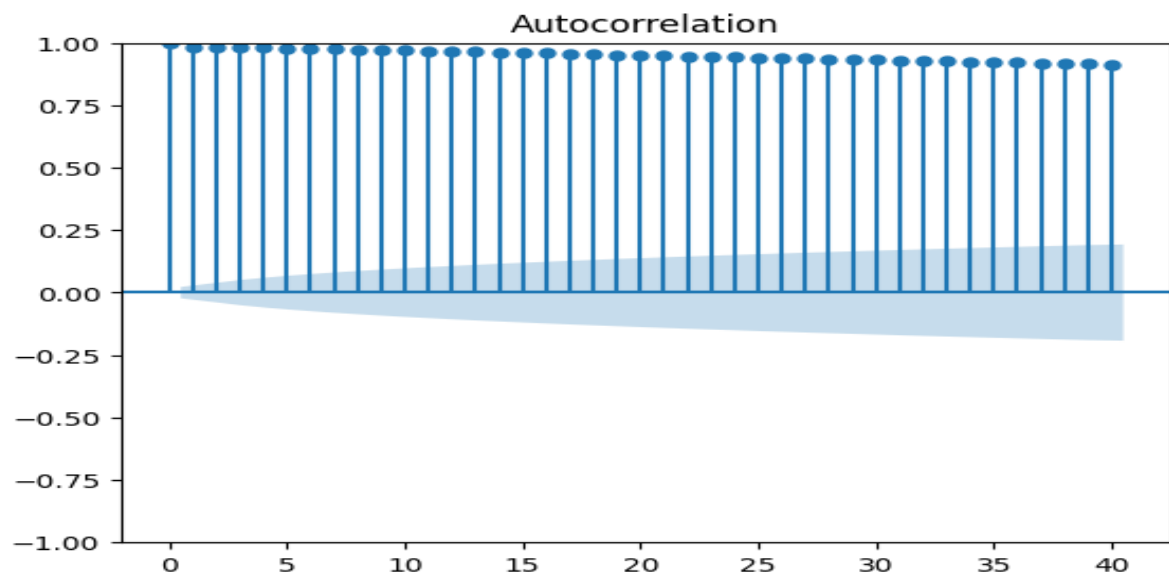


Context & Interpretation: Limited usefulness because only one weekday exists.

Python Script:

```
df.boxplot(column='red_corrected', by='weekday')
```

ACF Plot

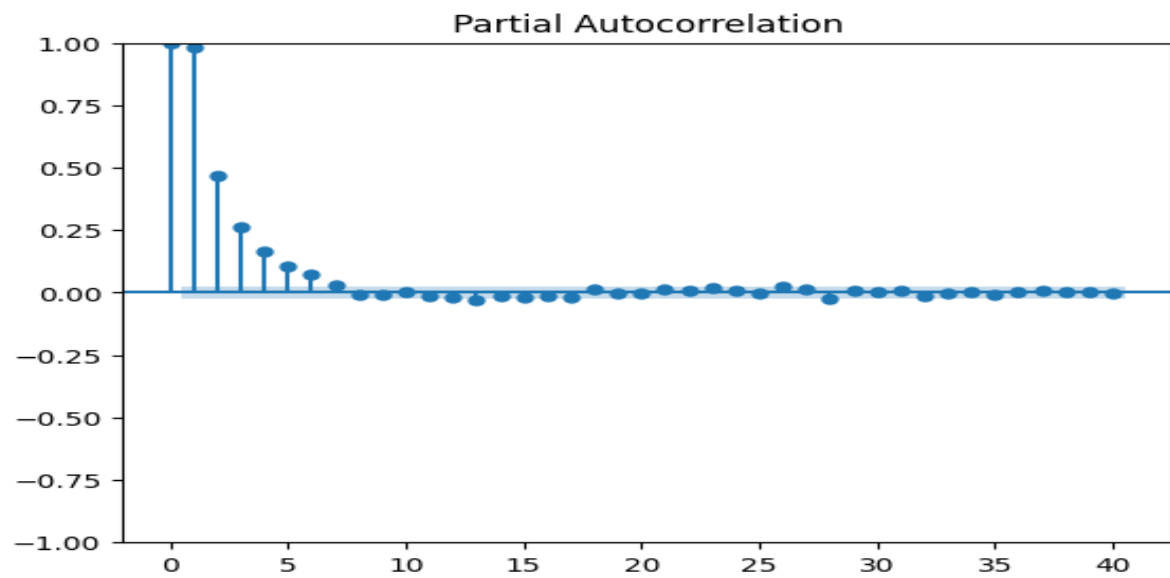


Context & Interpretation: Shows strong autocorrelation and temporal dependence.

Python Script:

```
plot_acf(df['red_corrected'], lags=40)
```


PACF Plot



Context & Interpretation: Shows strongest direct influence at initial lags.

Python Script:

```
plot_pacf(df['red_corrected'], lags=40)
```